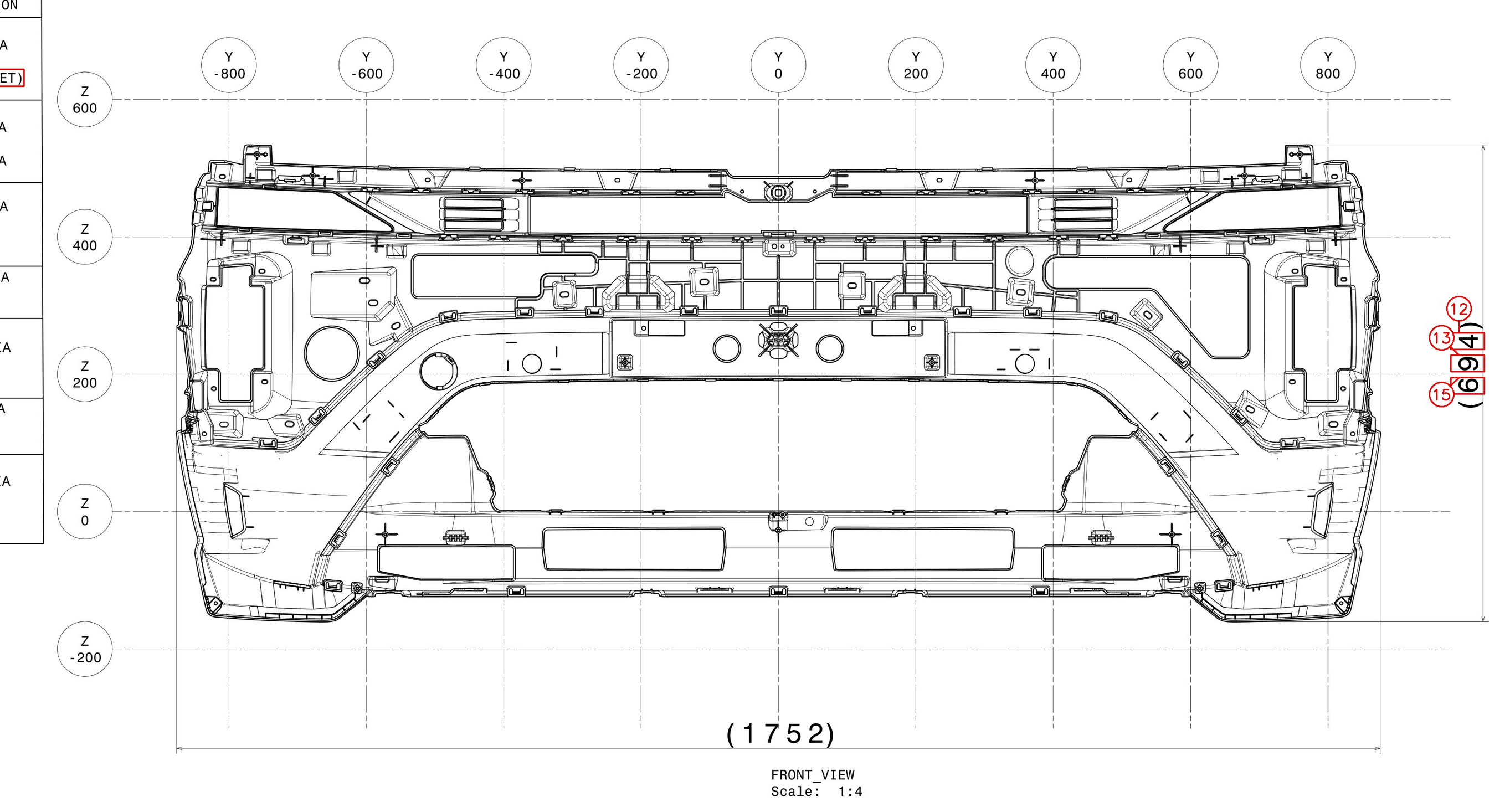
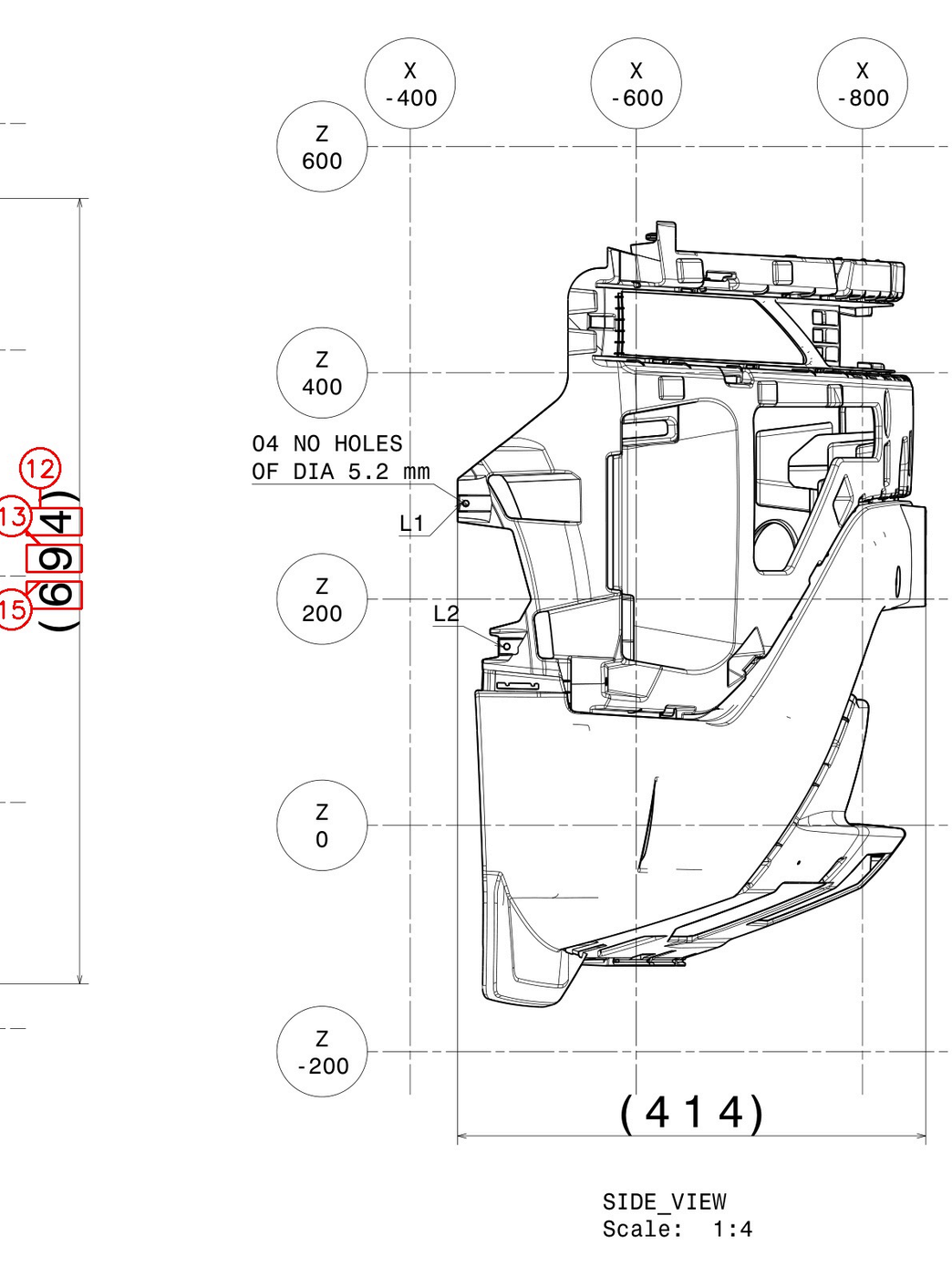
DATUM DEFINATION

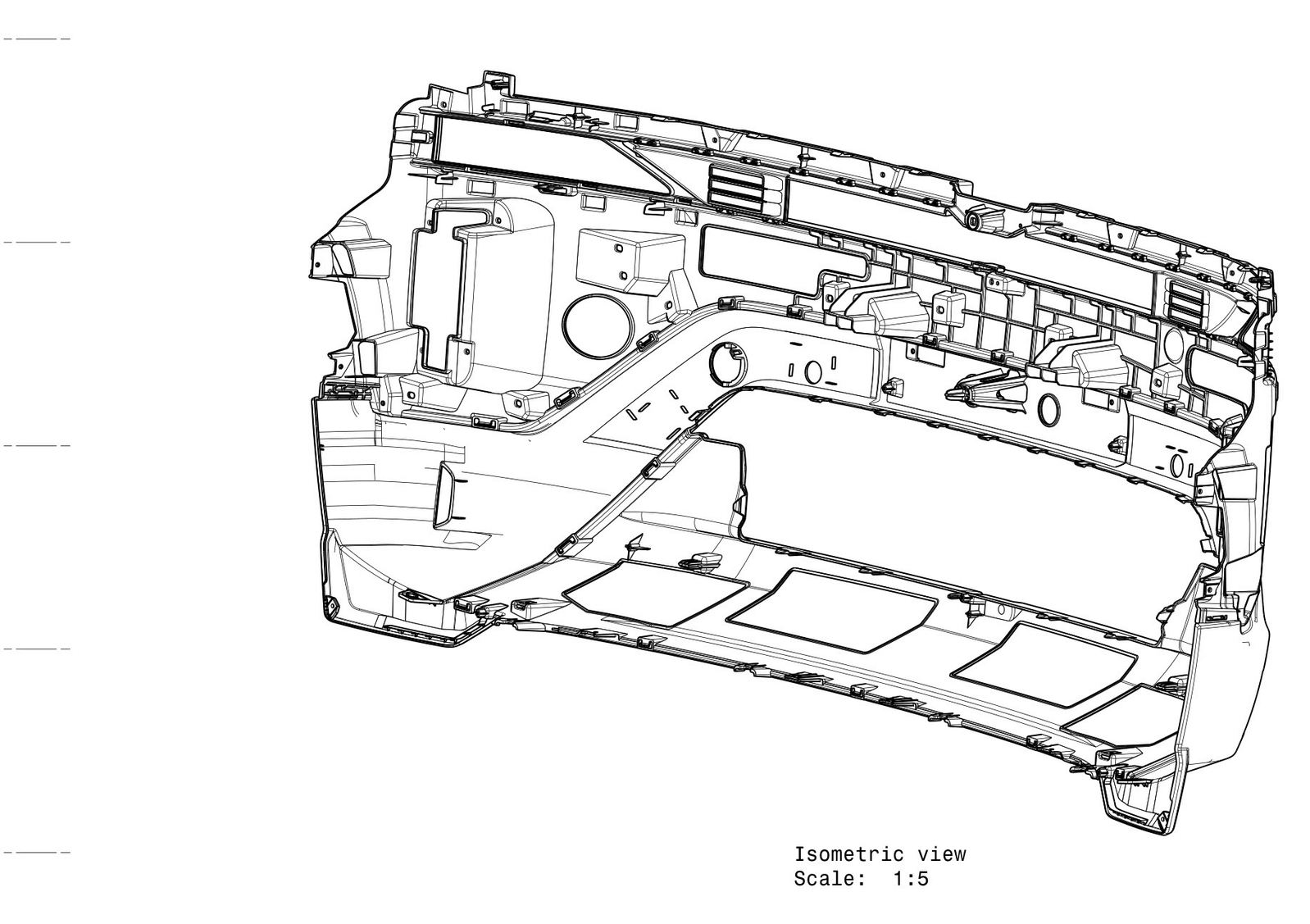
DATUM	DIRECTION	INTERACTION
A1	F/A	LOWER FASCIA WITH BIW(T-BRACKET)
B1-B2	C/C	
C1	U/D	
AA1 - AA22	F/A	LOWER FASCIA WITH UPPER FASCIA
BB1-BB13	C/C	
CC1 - CC18	U/D	
G1 - G13	F/A	LOWER FASCIA WITH LOWER GRILL
H1 - H2	C/C	
J1 - J11	U/D	
L1 - L6	F/A	LOWER FASCIA WITH SKID PLATE
M1 - M4	C/C	
P1 - P5	F/A	LOWER FASCIA WITH TOW HOOK
Q1	C/C	
R1 - R2	U/D	
S1 - S4	C/C	LOWER FASCIA WITH TOP GRILL
U1 - U10	U/D	
X1 - X6	F/A	LOWER FASCIA WITH AERO DUCT
Y1 - Y14	C/C	
Z1 - Z2	U/D	


Y -800
Y -600
Y -400
Y -200
Y 0
Y 200
Y 400
Y 600
Y 800
Z 600
Z 400
Z 200
Z 0
Z -200
(694)
(1752)

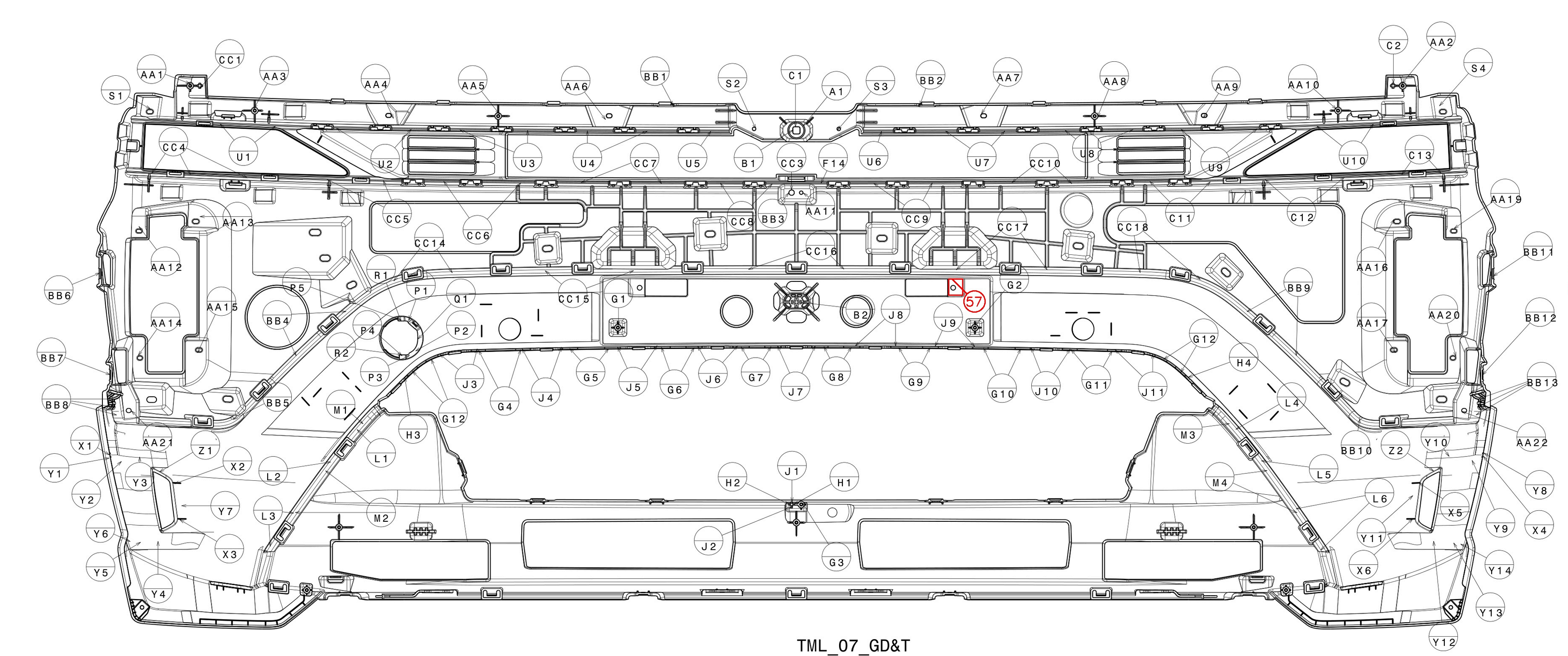
FRONT_VIEW
Scale: 1:4


X -400
X -600
X -800
Z 600
Z 400
Z 200
Z 0
Z -200
04 NO HOLES
OF DIA 5.2 mm
L1
L2
(414)

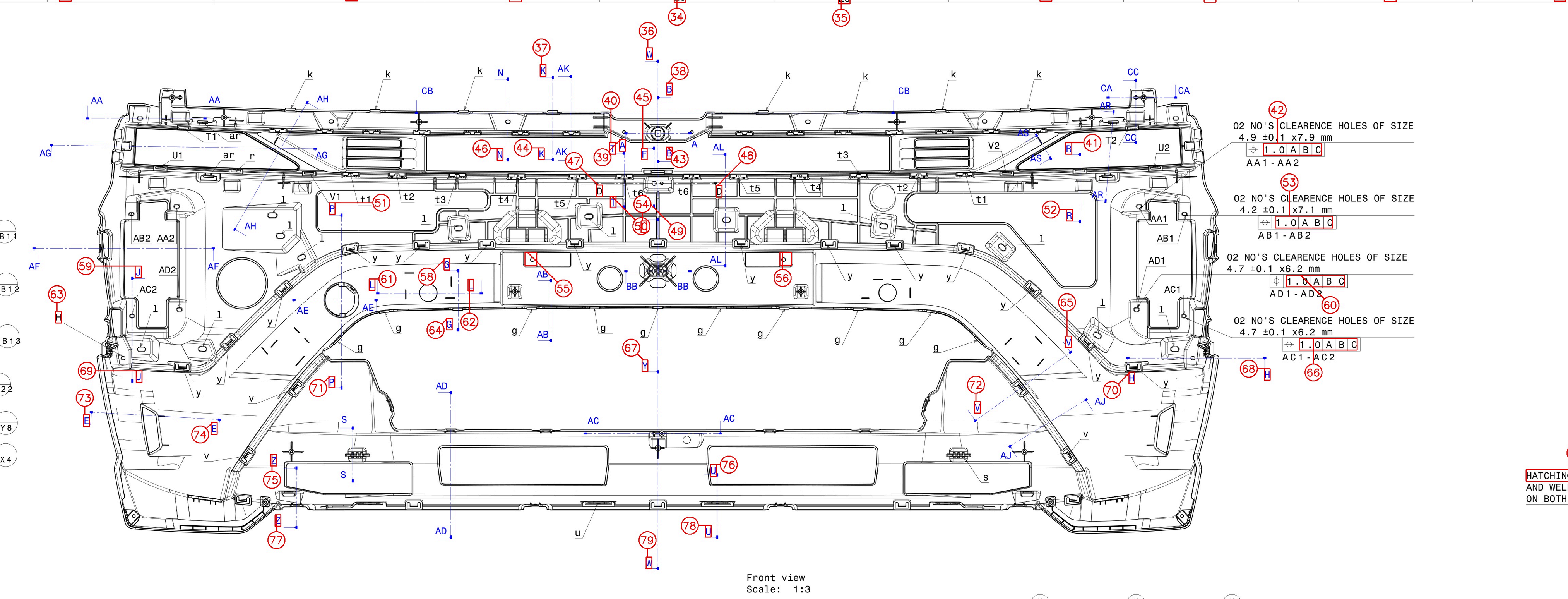
SIDE_VIEW
Scale: 1:4



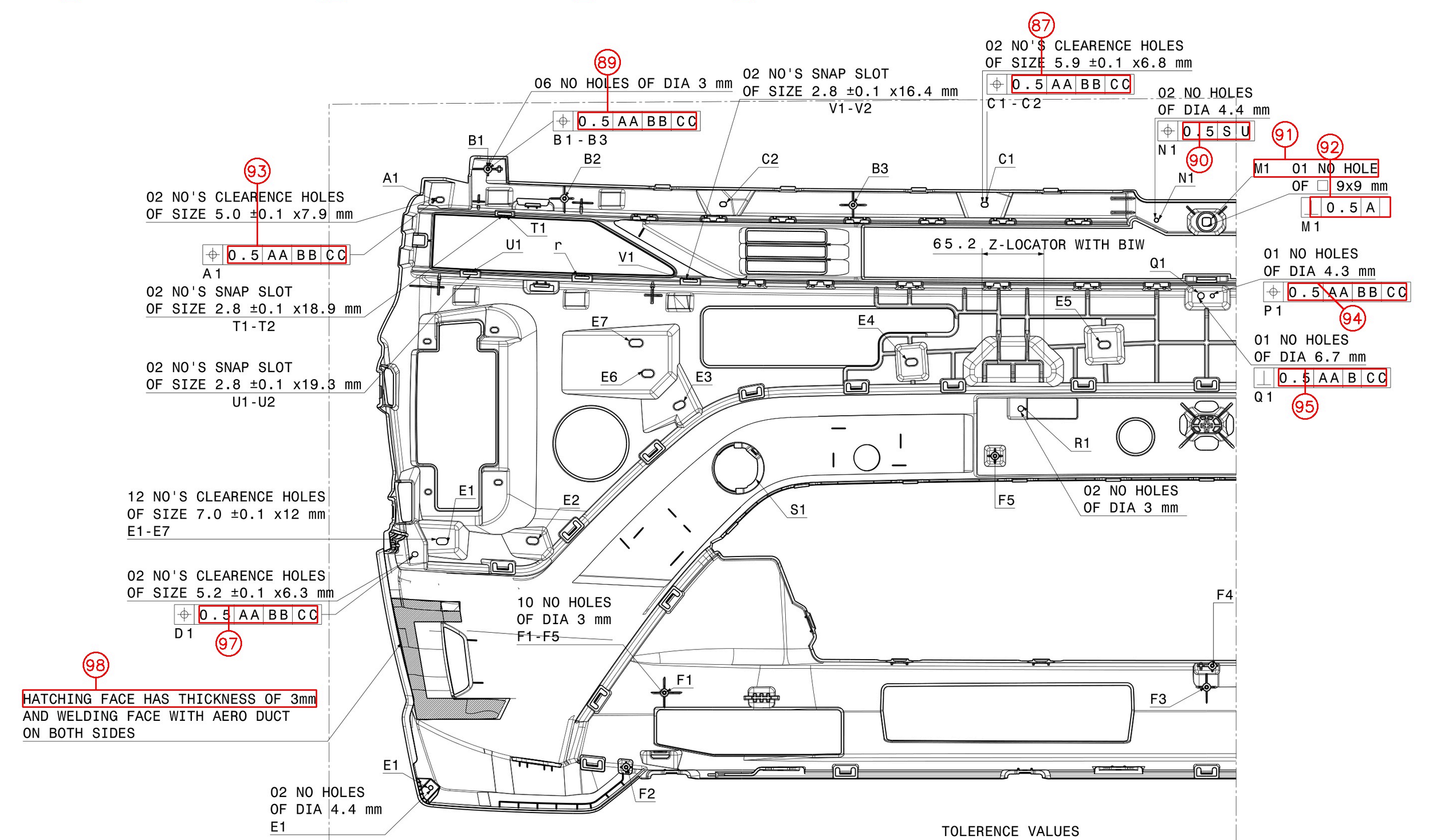
Isometric view
Scale: 1:5


AA1
CC1
AA3
AA4
AA5
AA6
BB1
S2
C1
S3
BB2
AA7
AA8
AA9
AA10
C2
AA2
S4
S1
U1
U2
U3
U4
U5
U6
U7
U8
U9
U10
B1
B2
A1
CC3
CC4
CC5
CC6
CC7
CC8
CC9
CC10
CC11
CC12
CC13
CC14
CC15
CC16
CC17
CC18
G1
G2
G3
G4
G5
G6
G7
G8
G9
G10
G11
G12
H1
H2
J1
J2
J3
J4
J5
J6
J7
J8
J9
J10
J11
P1
P2
P3
P4
P5
Q1
R1
R2
L1
L2
L3
L4
L5
L6
M1
M2
M3
M4
X1
X2
X3
X4
X5
X6
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
Y9
Y10
Y11
Y12
Y13
Y14
Z1
Z2

TML_07_GD&T
Scale: 1:3


02 NO'S CLEARENCE HOLES OF SIZE
4.9 ±0.1 x7.9 mm
1.0 A B C
AA1-AA2
02 NO'S CLEARENCE HOLES OF SIZE
4.2 ±0.1 x7.1 mm
1.0 A B C
AB1-AB2
02 NO'S CLEARENCE HOLES OF SIZE
4.7 ±0.1 x6.2 mm
1.0 A B C
AD1-AD2
02 NO'S CLEARENCE HOLES OF SIZE
4.7 ±0.1 x6.2 mm
1.0 A B C
AC1-AC2

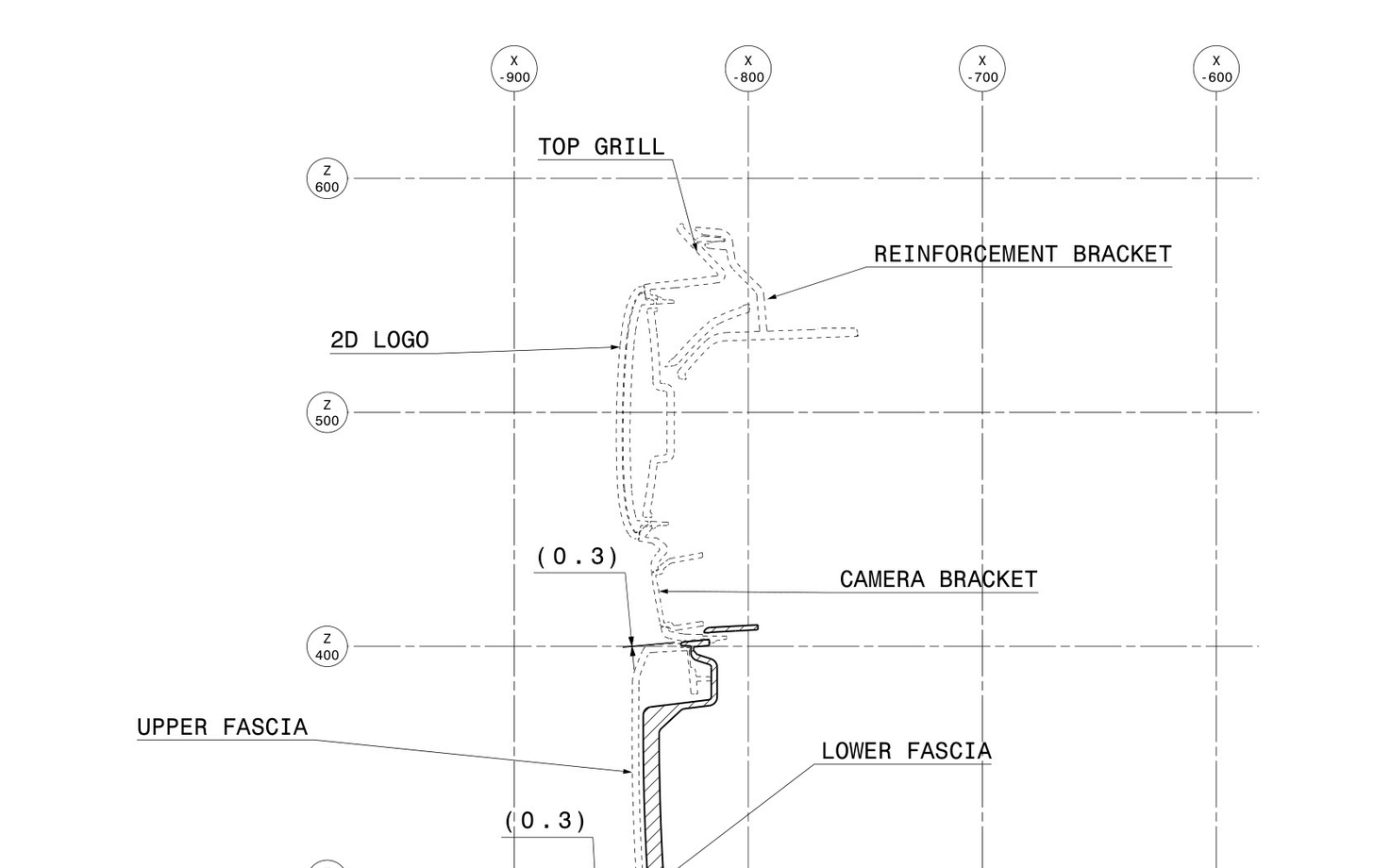
Front view
Scale: 1:3

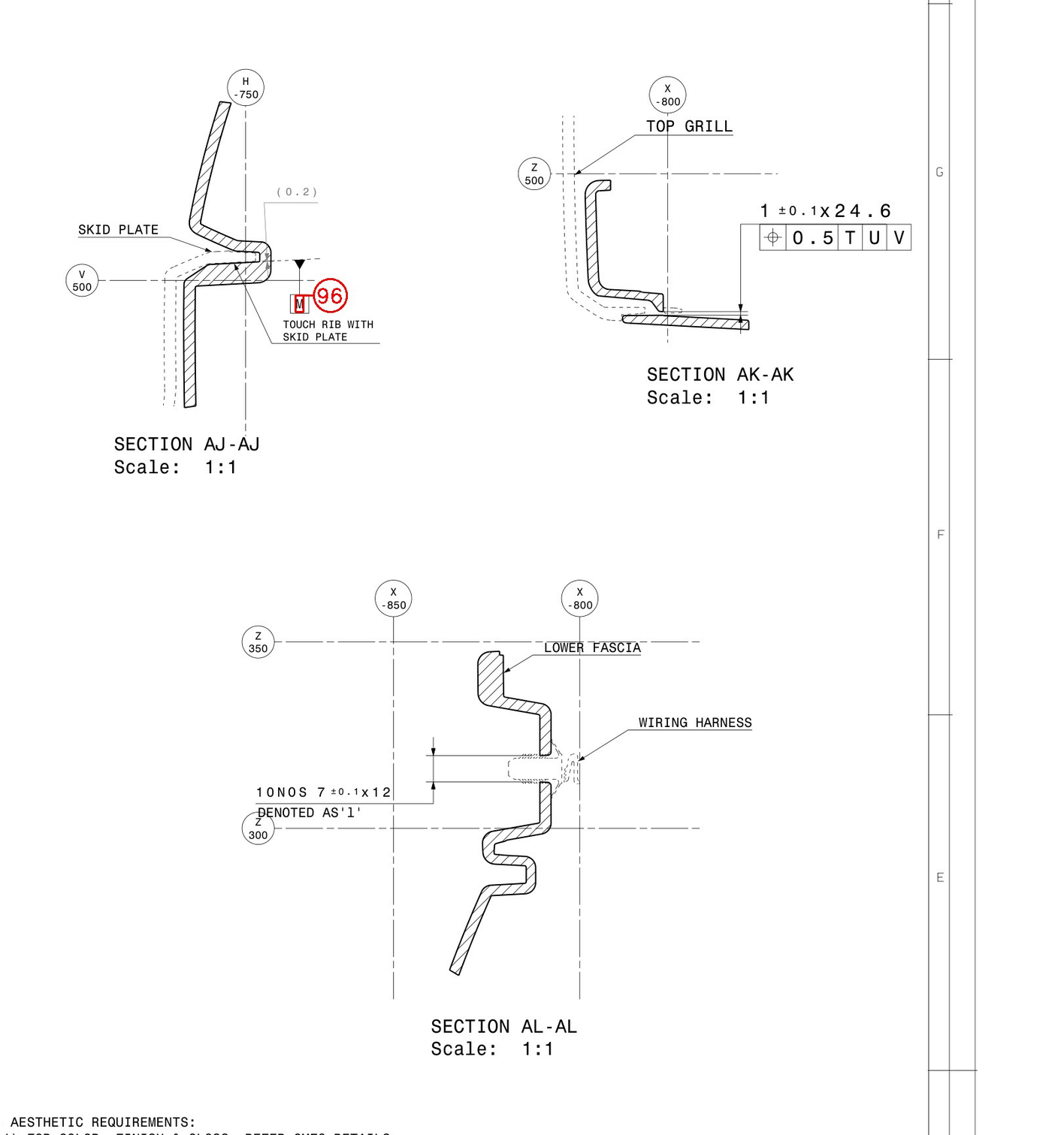

06 NO HOLES OF DIA 3 mm
0.5 AA BB CC
B1-B3
02 NO'S SNAP SLOT
OF SIZE 2.8 ±0.1 x16.4 mm
V1-V2
02 NO'S CLEARENCE HOLES
OF SIZE 5.9 ±0.1 x6.8 mm
0.5 AA BB CC
C1-C2
02 NO HOLES
OF DIA 4.4 mm
0.5 S U
N1
M1 01 NO HOLE
OF □9x9 mm
0.5 A
02 NO'S CLEARENCE HOLES
OF SIZE 5.0 ±0.1 x7.9 mm
0.5 AA BB CC
A1
02 NO'S SNAP SLOT
OF SIZE 2.8 ±0.1 x18.9 mm
T1-T2
02 NO'S SNAP SLOT
OF SIZE 2.8 ±0.1 x19.3 mm
U1-U2
65.2 Z-LOCATOR WITH BIW
01 NO HOLES
OF DIA 4.3 mm
0.5 AA BB CC
P1
01 NO HOLES
OF DIA 6.7 mm
0.5 AA B CC
Q1
02 NO HOLES
OF DIA 3 mm
12 NO'S CLEARENCE HOLES
OF SIZE 7.0 ±0.1 x12 mm
E1-E7
02 NO'S CLEARENCE HOLES
OF SIZE 5.2 ±0.1 x6.3 mm
0.5 AA BB CC
D1
10 NO HOLES
OF DIA 3 mm
F1-F5
HATCHING FACE HAS THICKNESS OF 3mm
AND WELDING FACE WITH AERO DUCT
ON BOTH SIDES
02 NO HOLES
OF DIA 4.4 mm
E1

TOLERENCE VALUES
Scale: 1:3

NOTE:
1)QUANTITY OF HOLES REPRESENTED AS PER SYMMETRY PART
2)E6-E7 AREA AND S1 AREA SHOWN ARE NON SYMMETRY

SR. NO.	CHANGE NOTICE NO.	CAD REV.	REVISION RECORD	AFF. FIELD	DATE	APP.
00		E	NEW RELEASE		2024/11/28	TSG
01		E	DRAWING SEC & DIMENTION UPDATED		2025/01/08	TSG
02		E	DIMENSION UPDATE		2025/02/12	TSG
03		E.2	DIMENSION UPDATE		2025/03/04	TSG

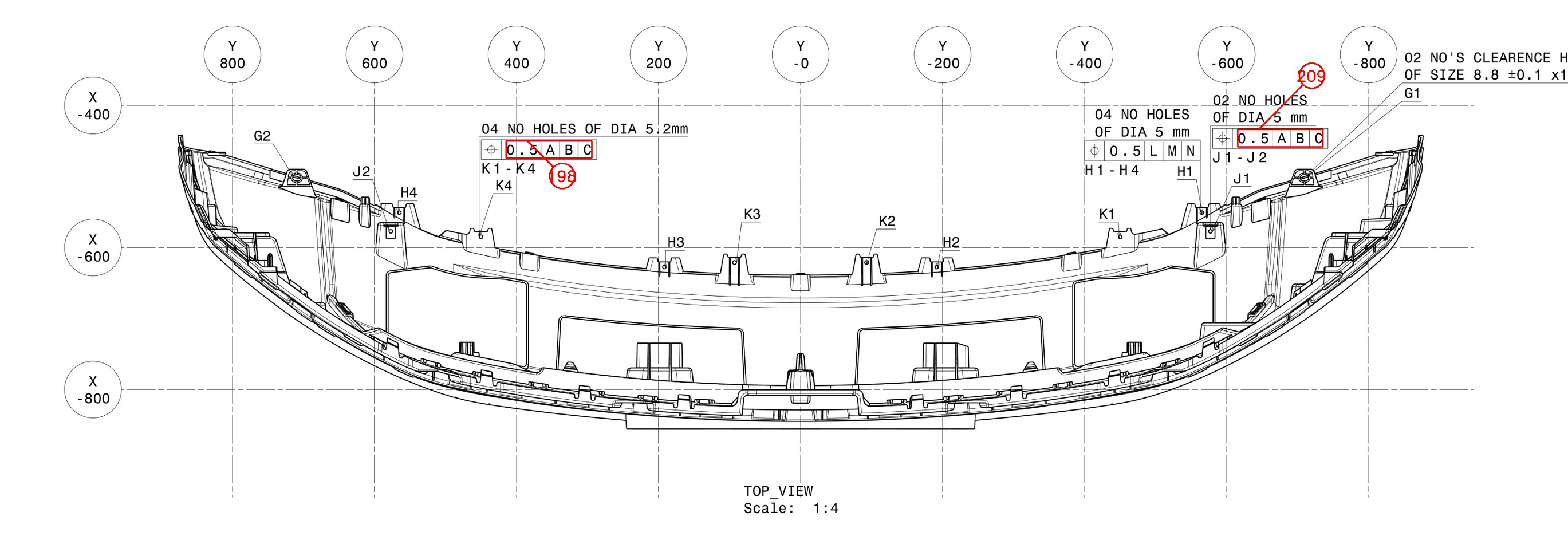

TOP GRILL
REINFORCEMENT BRACKET
2D LOGO
CAMERA BRACKET
UPPER FASCIA
LOWER FASCIA
(0.3)


SKID PLATE
TOUCH RIB WITH SKID PLATE
TOP GRILL
1 ±0.1 x24.6
0.5 T U V
LOWER FASCIA
WIRING HARNESS
10NOS 7 ±0.1 x12
DENOTED AS'l'

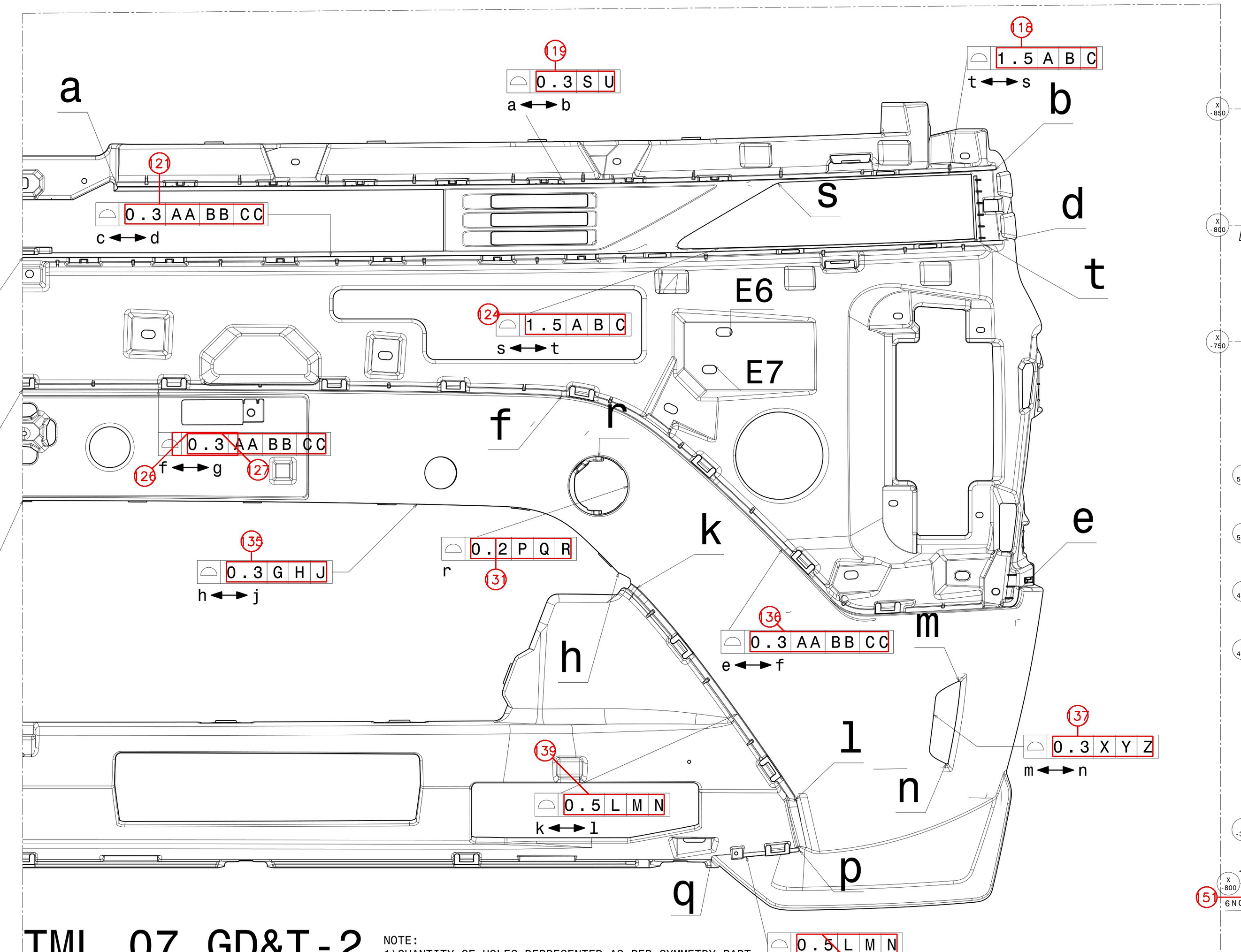
SECTION AJ-AJ
Scale: 1:1

SECTION AK-AK
Scale: 1:1

SECTION AL-AL
Scale: 1:1


04 NO HOLES OF DIA 5.2mm
0.5 A B C
K1-K4
04 NO HOLES
OF DIA 5 mm
0.5 L M N
H1-H4
02 NO HOLES
OF DIA 5 mm
0.5 A B C
J1-J2
02 NO'S CLEARENCE HOLES
OF SIZE 8.8 ±0.1 x12 mm
G1
G2
H1
H2
H3
H4
J1
J2
K1
K2
K3
K4
X -400
X -600
X -800

TOP_VIEW
Scale: 1:4


a
b
c
d
e
f
g
h
j
k
l
m
n
p
q
r
s
t
E6
E7
0.3 S U
a ↔ b
1.5 A B C
t ↔ s
0.3 AA BB CC
c ↔ d
1.5 A B C
s ↔ t
0.3 AA BB CC
f ↔ g
0.2 P Q R
r
0.3 G H J
h ↔ j
0.3 AA BB CC
e ↔ f
0.3 X Y Z
m ↔ n
0.5 L M N
k ↔ l
0.5 L M N
p ↔ q

TML_07_GD&T-2
Scale: 1:2

NOTE:
1)QUANTITY OF HOLES REPRESENTED AS PER SYMMETRY PART
2)E6-E7 AREA AND 'r' AREA SHOWN ARE NON SYMMETRY

2D LOGO
TOP GRILL
LOWER FASCIA
T-BRACKET
REINFORCEMENT BRACKET
FEM BIW
ALL LOCATORS MAY NEED TO FINE TUNE AFTER FITMENT & TRIAL. PLEASE PLAN INSERT ACCORDINGLY.
UPPER FASCIA
AERO DUCT
3M Acrylic Foam Tape PX5011
FPAS BRACKET
0.3 AA BB CC
GAP FACE WITH UPPER FASCIA
TOW HOOK COVER
FOG LAMP BEZEL
2NO 4.2 ±0.1 x7.1
1.0 A B C
DENOTED AS'AB'
2NOS 8.4 ±0.1 x20
0.5 A B C
DENOTED AS's'
Φ5 ±0.1
0.5 L M N
11NOS 3.5 ±0.1
DENOTED AS'g'
0.3 G H J
GAP FACE WITH LOWER GRILL
309.2 ±0.1
0.1 AA
8 NO 4.8 ±0.1 x15
0.5 A B C
DENOTED AS'k'
ALL EIGHT SNAP FACES MAY NEED TO FINE TUNE AFTER FITMENT & TRIAL. PLEASE PLAN INSERT ACCORDINGLY.
0.3 S U
GAP FACE WITH UPPER GRILL
2NOS 6.3 ±0.1 x5.2
0.5 A B C
DENOTED AS'h'
0.3 D E F
GAP FACE WITH UPPER FASCIA
0.3 X Y Z
FLUSH FACE WITH UPPER FASCIA
15NOS 2.2 ±0.1
0.5 AA BB CC
DENOTED AS'y'
4NOS 2.2 ±0.1
DENOTED AS'v'
SKID PLATE
TABLE FOR SECTION T-T
Length(L)
1.4 ±0.1
1.3 ±0.1
1.3 ±0.1
1.3 ±0.1
1.4 ±0.1
1.4 ±0.1
DENOTED AS't'
LOWER GRILL
WHEEL ARCH CLADDING
1.5 A B C
GAP INTERFACE WITH WAC
1.0 A B C
FLUSH INTERFACE WITH WAC
5.2 ±0.1
0.5 A B C
GAP FACE WITH HEAD LAMP
HEAD LAMP
1.5 A B C
GAP INTERFACE WITH HEAD LAMP
2.8 ±0.1
DENOTED AS'r'
LOWER STIFFNER
2NOS Φ5.2 ±0.1
0.5 AA BB CC
DENOTED AS'u'
RADAR COVER
△1.9
NEED TO FINE TUNE AFTER FITMENT & TRIAL. PLEASE PLAN INSERT ACCORDINGLY.
0.2 P Q R
GAP FACE WITH TOW HOOK COVER
0.3 P Q R
FLUSH FACE WITH TOW HOOK COVER
TOUCH RIB THICKNESS ON BOTH SIDES IN X-AXIS
6NOS 4 ±0.1
0.5 L M N
△1.7
6.2 ±0.1
0.5 G H J
3 ±0.1
0.1 H
3 ±0.1
0.5 A B C

SECTION A-A
Scale: 1:1

SECTION CA-CA
Scale: 1:1

SECTION CC-CC
Scale: 1:1

SECTION D-D
Scale: 1:1

SECTION E-E
Scale: 1:1

SECTION L-L
Scale: 1:1

SECTION N-N
Scale: 1:1

SECTION AA-AA
Scale: 1:1

SECTION B-B
Scale: 1:1

SECTION BB-BB
Scale: 1:1

SECTION G-G
Scale: 1:1

NOTE:△ REFERS TO IMP DIM.

SECTION H-H
Scale: 1:1

SECTION P-P
Scale: 1:1

SECTION J-J
Scale: 1:1

SECTION S-S
Scale: 1:1

SECTION Z-Z
Scale: 1:1

SECTION CB-CB
Scale: 1:2

SECTION K-K
Scale: 1:1

NOTE:△ REFERS TO IMP DIM.

SECTION F-F
Scale: 1:1

SECTION Q-Q
Scale: 1:1

SECTION Y-Y
Scale: 1:1

SECTION V-V
Scale: 1:1

SECTION T-T
Scale: 1:1

SECTION AB-AB
Scale: 1:1

SECTION AF-AF
Scale: 1:1

SECTION AH-AH
Scale: 1:1

SECTION AR-AR
Scale: 1:2

NOTE:△ REFERS TO IMP DIM.

SECTION W-W
Scale: 1:2

SECTION AG-AG
Scale: 1:1

SECTION R-R
Scale: 1:1

SECTION AS-AS
Scale: 1:2

SECTION XX-XX

NOTE:△ REFERS TO IMP DIM.

SECTION AE-AE
Scale: 1:1

SECTION U-U
Scale: 1:1

SECTION AD-AD
Scale: 1:1

SECTION AC-AC
Scale: 1:1

A) AESTHETIC REQUIREMENTS:
1) FOR COLOR, FINISH & GLOSS, REFER CMFG DETAILS
2) PART SHOULD HAVE AESTHETIC APPROVAL REPORT (AAR)

B) MATERIAL REQUIREMENTS:
1) MATERIAL TO BE APPROVED BY TML MATERIAL TEAM

C) QUALITY REQUIREMENTS:
1) GENERAL WALL THK: 2.8 MM
2) INJECTION MOLDED PART DIMENSIONAL TOLERANCES TO BE AS PER TS:12121
3) APPEARANCE PART SHOULD MEET TS:11417
4) COMPONENT SHOULD BE FREE FROM VISUAL DEFECTS LIKE SINK MARKS, FLASHES, FLOW LINES, WHITE MARKS, SHRINK MARKS ETC
5) MATERIAL IDENTIFICATION MARKING ON PLASTIC COMPONENTS AS PER TS 11418. E.G. >ABS<
6) PART TO BE PROVE ON CHECKING FIXTURE WITHOUT FLEXING
7) CTQ PARAMETERS TO BE SIGNED OFF WITH TML ERC
8) CRITICAL TO QUALITY DIMENSIONS PIST TO BE ENSURE
9) SAMPLE TO BE APPROVED BY TML ERC QA TEAM PRIOR TO TESTING. SAMPLE TO BE PROVIDED WITH INSPECTION & MATERIAL REPORT WHILE TESTING
10) GAP & FLUSHNESS TO BE ACHIEVED AS PER PERCEIVED QUALITY GAP PLAN

D) TESTING & HOMOLOGATION REQUIREMENTS:
1) ENVIRONMENTAL TESTING AS PER TS: 75006
2) COMPONENT TO MEET CHEMICAL RESISTANCE TEST AS PER TS 76065
3) SCRATCH RESISTANCE FORCE MINIMUM IN FOR ONSET OF WHITENING AS PER TS 75014
4) PART MATERIAL WEIGHT DATA SUBMISSION AND HAZARDOUS MATERIAL DECLARATION AS PER TS:11420
5) COMPLIANCE WITH RRR AND ELV DIRECTIVE (2000/53/EC AND 2005/64/EC)
6) PART MATERIAL WEIGHT DATA SUBMISSION AND HAZARDOUS MATERIAL DECLARATION AS PER TS:11420, SUPPLIER TO SUBMIT/DECLARE THE SAME FOR ASSEMBLY LEVEL & PART LEVEL
7) WATER IMMERSION TEST: NO LOSS OF ADHESION, NO BLISTER ON ANY OTHER VISUAL DEFORMATION
8) COMPONENT TO MEET BSR TEST AS PER TST/TS/WI/163
9) FOR MORE DETAILS PLEASE GO THROUGH SIGNED OFF DVP/CTR

E) GENERAL REQUIREMENTS:
1) UNSPECIFIED RADII 0.5 R AND DRAFT 1°
2) GENERAL SEEN SURFACE WITH BLENDED UNIFORMITY
3) ADDITIONAL RIBS LOCATION AND THICKNESS TO BE DECIDED BY SUPPLIER FOR PART STRENGTH, DIMENSIONS CONTROL AVOIDING SINK MARKS
4) THERE HAS TO BE FORMAL DATA APPROVAL BY TATA MOTORS BEFORE ANY TOOL CUTTING
5) FOR DETAILED TOOL SAFE CONDITION, REFER SURFACES MENTIONED IN MAGENTA COLOR IN 3D & SAFE STEEL CONDITION SHOULD MAINTAIN IN TOOL FOR FURTHER DEVELOPMENT
6) REQUIRED PART FITMENT/SNAP FITMENT/GAP & FLUSH CONDITION TO BE ACHIEVED WITH CHILD PARTS AT THE SUPPLIER END IN ASSEMBLY CONDITION.
7) FOR ANY DOUBT,SUPPLIER SHOULD ASK BEFORE TOOL DESIGN
8) IF ANY DOUBT SUPPLIER SHOULD ASK FOR 'A', 'B' & 'C' ZONE CLARIFICATION
9) CMM & WHITE LIGHT SCANNING REPORT TO BE SUBMITTED W.R.T 3D DATA FOR T1,T2 & T3 PHYSICAL SAMPLES
10) WHEREVER FEASIBLE, REPLACEABLE INSERTS TO BE USED AT HOLE / SLOT AREA.THIS HAS TO BE DISCUSSED WITH TML WHILE TOOL DESIGN.
11) FOR ALL MATING AND NON-MATING SURFACES, REFER 3D CAD DATA
12) RETENTION TORQUE VALUE TO BE WITHIN 10% OF APPLICATION TORQUE VALUE
13) PARTING LINE AND ACCEPTANCE CRITERIA, PLS REFER PQ GUIDELINE

F) PART SPECIFIC REQUIREMENTS:
1) WELDED BRACKET ADHESION STRENGTH WITH FRONT BUMPER TO BE 20 KGFM
2) ENDURANCE OF SNAPS :- NO DAMAGE OF SNAPS AFTER 5 TIMES FITMENT AND REMOVAL
3) EXTERIOR PROJECTION AS PER AIS120/ ECE R26
4) VIBRATION TEST AS PER AIS 006
5) PEEL OFF FORCE TEST:- PART SHOULD NOT FALL OUT AFTER APPLYING 1000 TIMES THE WEIGHT OF THE COMPONENT AFTER 24 HRS OF PASTING THE PART ON CLEANED BASE SURFACE. PREFERRED LOCATION IS AS PER FINAL VEHICLE INTENT
6) CHILD LEVEL PARTS ASSEMBLY TORQUES TO BE ENSURE IN ANNOTATION SECTION

<---DRF--->

A	B	C

GENERAL TOLERANCE UNLESS OTHERWISE SPECIFIED

1.PART SYMMETRICAL ABOUT Y=0 UNLESS OTHERWISE SPECIFIED.
2.ALL SIZE AND FORM TOLERANCES TO BE VERIFIED PRIOR TO LOADING THE PART ON GAUGE.
3.ALL PARTS TO BE RESTRAINED ON DATUMS UNLESS OTHERWISE SPECIFIED.
4.CAD DATA IS MASTER.
5.FULL SIZE PIN WITH +0/-0.01 TO BE USED ON GAUGE AND FIXTURE.

ALL NON MATCHING SURFACES PROFILE	⌓ 3.0 DRF
ALL NON-KEY CIRCULAR FEATURES	⌖ Φ3.0 DRF
ALL NON-KEY TRIM EDGES	⌓ 3.0 DRF
ALL NON-KEY NON CIRCULAR FEATURES	⌖ Φ3.0 DRF
UNSPECIFIED HOLE SIZE TOLERANCE	+0.2/-0.0

ITEM	PART NO.	PART NAME	QTY.	MATERIAL	COLOUR	REMARKS

CUS. PART NO :- 547688506303	MASS: - 3918 gm	DRAWN HARSHA
CUS. PART NAME :- FR BUMPER INTEGRATED FASCIA WITH FPAS 2SVS	MATERIAL: - PP EPDM TD13	CHECKED SUJIT
MATE PART NAME :- FR BUMPER INTEGRATED FASCIA WITH FPAS 2SVS	COLOUR: - AS PAINTED	APPROVED TULSIDAS
MATE PART NO :- TBD	SURFACE FINISH: - PIANO BLACK	
	SHEET NO.: - 1 OF 1	
MATE Motherson Automotive Technologies & Engineering Gat no. 150,Tal- Khed. Dist. Pune.410501	SCALE SEE DRG.	A0

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ALL DIMENSIONS ARE IN mm, DO NOT SCALE THE DRG. IF IN DOUBT PLEASE ASK.